E-Commerce Analytics Report – Olist Dataset

1. Cover Page

Project Title: Olist E-Commerce Analytics Dashboard

Sector: E-Commerce / Retail Analytics

Institute: Newton School of Technology

GitHub: https://github.com/GitSid-glitch/Capstone_Tableau.git

Tableau Dashboard: https://public.tableau.com/app/profile/vinay.sharma7499/viz/OlistE-CommerceAnalyticsDashboard_17773877162720/Story1

Submission Date: 28-04-2026

Team Members

Vinay Sharma
Vansh Jain
Sumit Kumar
Siddharth Kumar Shukla
Sumit Kumar
Ashwin Tiwari

2. Executive Summary

This project presents a comprehensive analysis of the Olist e-commerce dataset to evaluate business performance,
customer satisfaction, and logistics efficiency. The study integrates Python-based data preprocessing with Tableau
visual dashboards to generate actionable insights.

The analysis reveals that delivery delays are the most critical factor impacting customer ratings. Orders delivered on time
consistently receive higher satisfaction scores, while delays significantly increase negative feedback. Additionally, revenue
is heavily concentrated in a few states such as São Paulo, indicating a lack of geographic diversification.

Another key observation is the dominance of credit card payments, which introduces financial dependency risk. Based on these findings,
the report recommends improving logistics infrastructure, expanding into underperforming regions, and diversifying payment methods
to enhance business resilience and customer experience.

3. Sector & Business Context

The e-commerce sector has experienced rapid growth due to increased internet penetration and digital payment adoption.
However, this growth is accompanied by operational challenges such as inefficient logistics, high competition, and evolving
customer expectations.

Companies must ensure fast delivery, reliable service, and seamless user experience to remain competitive. Decision-makers rely
on data-driven insights to optimize supply chain operations and improve customer retention.
This project provides such insights
to help businesses make informed strategic decisions.

4. Problem Statement & Objectives

The primary problem addressed in this project is the imbalance between growing sales and declining customer satisfaction
due to delivery inefficiencies.

Objectives:
• Identify key revenue-driving regions and product categories
• Analyze delivery performance and delays
• Evaluate customer satisfaction using review scores
• Provide actionable business recommendations

Success is measured by the ability to generate insights that directly support operational improvements and strategic decisions.

5. Data Description

The dataset consists of multiple relational tables including orders, customers, sellers, products, payments, and reviews.
It covers over 100,000 transactions between 2016 and 2018.

Key fields include order timestamps, delivery dates, payment types, product categories, and customer ratings.
Limitations include missing values in delivery data, regional bias toward Brazil, and lack of real-time updates.

6. Data Cleaning & ETL Pipeline

Data preprocessing was performed using Python. Missing values were handled using imputation and removal techniques.
Timestamps were standardized, and new features such as delivery time and late delivery

indicators were created.

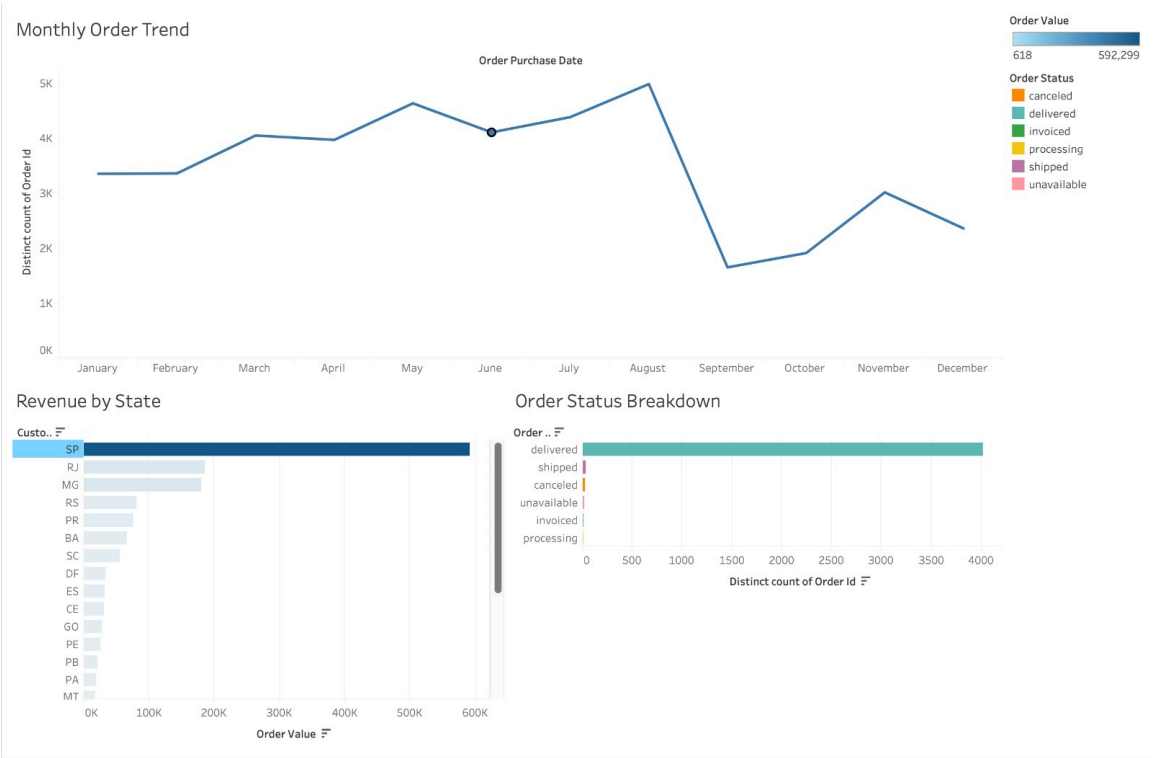
Outliers in payment values were removed to maintain data consistency. These steps ensured that the dataset was clean,
reliable, and suitable for analysis.

7. KPI Framework

Key performance indicators include revenue, order volume, average delivery time, customer rating, and late delivery rate.
These KPIs provide a comprehensive view of business performance, operational efficiency, and customer satisfaction.

Each KPI directly maps to business objectives, enabling stakeholders to track performance and identify areas for improvement.

8. Dashboard Analysis – Operations



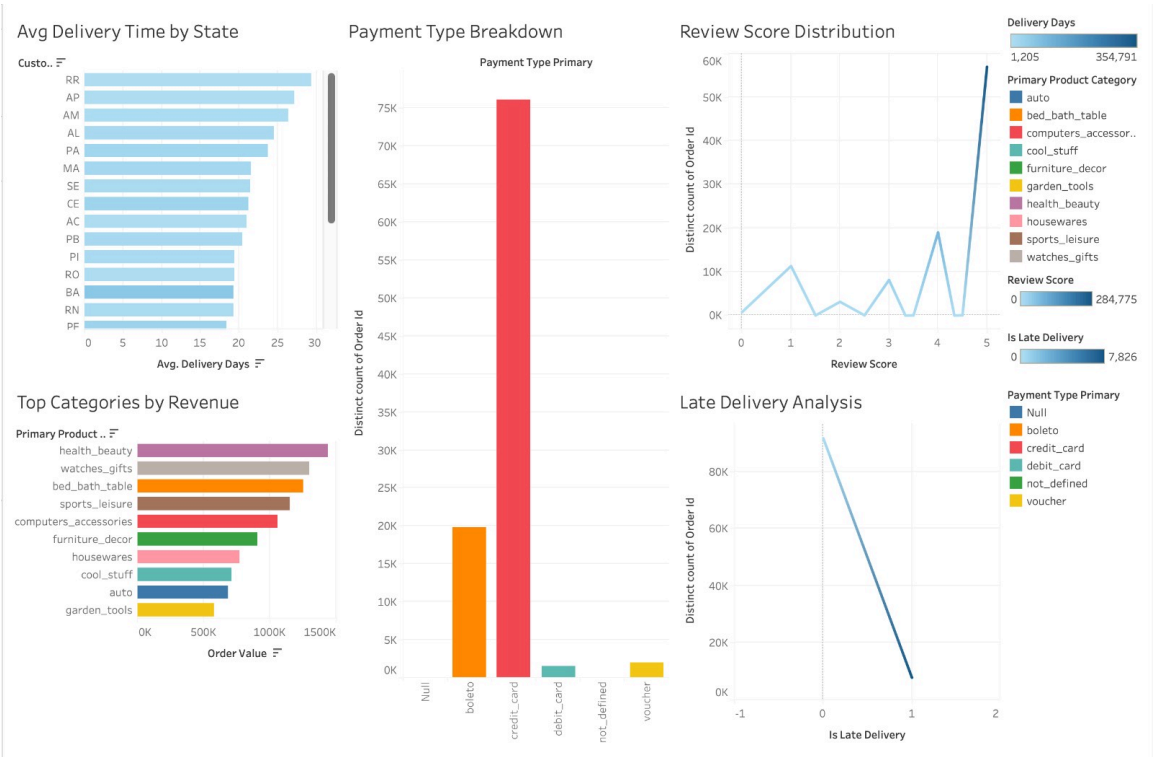
The operational dashboard highlights delivery inefficiencies across regions. States with longer delivery times show
significantly lower customer ratings, indicating a direct impact on satisfaction.

Payment analysis reveals that credit cards dominate transactions, creating dependency risk.
Product category analysis shows

that certain categories consistently generate higher revenue, indicating strong customer demand.

The relationship between late deliveries and negative reviews emphasizes the need for logistics optimization.

9. Dashboard Analysis – Revenue & Trends



Revenue analysis shows heavy concentration in São Paulo, indicating geographic imbalance. Monthly trends reveal
seasonal fluctuations, suggesting the need for demand forecasting.

Order status distribution indicates operational inefficiencies in cancellations and unavailable orders.
This highlights the importance of improving inventory and delivery processes.

10. Exploratory Data Analysis

EDA reveals a strong negative correlation between delivery time and customer ratings. Faster deliveries consistently
result in higher ratings, while delays significantly reduce satisfaction.

Revenue concentration in a few states increases risk. Category-wise analysis shows uneven

demand distribution,
indicating the need for diversification strategies to ensure sustainable growth.

11. Statistical Analysis

Customer segmentation analysis shows that high-value customers are concentrated in urban areas.
Improving service quality in these regions can significantly boost revenue.

Statistical models confirm that reducing delivery time leads to higher customer satisfaction and repeat purchases,
making logistics optimization a key investment area.

12. Tableau Dashboard Design

The Tableau dashboard provides both executive summaries and operational insights. Interactive filters allow users
to analyze data by region, category, and time period.

The design focuses on clarity and usability, ensuring that stakeholders can quickly interpret insights and make decisions.

13. Insights Summary

Delivery delays are the primary driver of dissatisfaction. Revenue concentration creates business risk, and payment
dependency limits flexibility. Seasonal demand fluctuations impact operations, and certain categories consistently outperform others.

14. Recommendations

Improve logistics in high-delay regions, diversify payment methods, expand geographically, and implement demand forecasting.
These actions will improve efficiency, reduce risk, and enhance customer satisfaction.

15. Conclusion

This project demonstrates the importance of data-driven decision-making in e-commerce. By addressing delivery inefficiencies
and expanding strategically, businesses can significantly improve performance and customer experience.

16. Contribution Matrix & Role Description

Vinay Sharma – Tableau Dashboard Development

Vinay Sharma was responsible for designing and developing the primary Tableau dashboard used in this project.
He structured both executive-level and operational views, ensuring that key metrics such as revenue, delivery time, and
customer satisfaction were clearly visualized. He implemented interactive filters for region, category, and time period
to allow dynamic exploration of data. His work focused on creating a user-friendly and visually intuitive dashboard that
supports effective decision-making. He also ensured consistency in design, layout, and storytelling across all dashboard views.

Sumit Kumar – Tableau Support & Visualization Enhancement

Sumit Kumar contributed to the Tableau development process by creating additional supporting visualizations
that complemented the main dashboard. His work ensured that multiple views provided deeper insights without redundancy.
He focused on refining charts, validating calculations, and improving visual clarity. He also helped test dashboard
interactivity and ensured that filters and parameters worked correctly across different views. His contribution enhanced
the overall usability and performance of the dashboard.

Siddharth Kumar Shukla – Data Cleaning & ETL Pipeline

Siddharth Kumar Shukla handled dataset selection, cleaning, and preprocessing using Python.
He designed the ETL pipeline, including handling missing values, converting timestamps, and standardizing formats.
He also created new features such as delivery time, late delivery flags, and aggregated metrics required for analysis.
His work ensured data quality, consistency, and readiness for further analysis. Additionally, he documented all
transformations and assumptions, making the process reproducible and aligned with project requirements.

Ashwin Tiwari – Exploratory Data Analysis & Statistical Analysis

Ashwin Tiwari conducted exploratory data analysis (EDA) and statistical analysis to identify patterns and trends
within the dataset. He analyzed relationships between variables such as delivery time and customer ratings, identifying key
factors impacting business performance. He also performed segmentation analysis to understand customer behavior and

distribution across regions. His work provided the analytical foundation for insights and recommendations presented in the report.

Vansh Jain – Report Writing & Documentation

Vansh Jain was responsible for preparing the final project report and documentation. He structured the report
according to academic guidelines and ensured clarity in presenting insights, analysis, and recommendations. He translated
technical findings into business-friendly language suitable for stakeholders. He also ensured that all sections were
logically connected and aligned with project objectives. His contribution ensured the report was comprehensive, well-organized,
and ready for submission.

17. Contribution Matrix

This section clearly documents each team member's contribution across all project phases.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing
Vinay Sharma					✓	
Vansh Jain						✓
Sumit Kumar					✓	
Siddharth Kumar Shukla	✓	✓				
Ashwin Tiwari			✓	✓		